Daniel Mitroshkov | dmkov15@cs.washington.edu | danielmkv.github.io

Computer Science undergraduate building AI research software, LLM evaluation tools, and automated cloud data pipelines.

EDUCATION

University of Washington | Seattle, WA |
Bachelor of Science in Computer Science | Expected **June 2028**

- **Graduate Coursework:** Deep Learning, Human/AI Interaction
- Additional Coursework: Modern Concurrency, Systems programming, Data Structures and Algorithms, Hardware & Software Interface, Linear Algebra, Probability and Statistics in Computing

SKILLS

- Programming: Rust, Python, C++, Java, SQL, C#, JavaScript, TypeScript, HTML/CSS
- AI & Data: PyTorch, LangChain, pandas, DuckDB, Jupyter
- Engineering & Cloud: AWS (Lambda, S3, EventBridge, ECR, CloudFront), Docker, Git, pytest, JUnit, Node.js

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EXPERIENCE

Undergraduate Researcher | Interactive Data Lab | Seattle, WA | December 2025 - Present
Research Experiences for Undergraduates (REU) Intern | June - September 2026

- Developing ADIEU, an LLM-based extension of prior algorithmic simulation research, adding agent simulation and evaluation capabilities
- Building modular research software and data-processing tools with Python and DuckDB to support reproducible experiments
- Evaluated 5+ LLM-powered data-analysis workflows and reviewed 50+ AI/HCI papers to inform feature development
- Collaborating with graduate researchers on software design, debugging, validation, and code review

Teaching Assistant | Paul G. Allen School of Computer Science and Engineering | Seattle, WA | September 2025 - Present

- Coordinating with CSE 394 course staff to help students begin conducting computer science research
- Previously supported 25+ students quarterly in CSE 197Y, a workshop for UW's introductory programming series
    - Led 3 hrs/week of one-on-one tutoring in programming fundamentals and problem solving
    - Developed **80+** JUnit test cases, refined course materials, and co-led staff meetings to align grading and support

IT Technician Intern | Richland School District | Richland, WA | June 2022 - July 2024

- Resolved **500+** help-desk tickets across **21 schools** while supporting a fleet of **6,000+** devices
- Improved Excel and Google Sheets inventory tools tracking **200+** shipments

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PROJECTS

contraiλ | AWS Hacks, 4th Place | May 2026

- Built an event-driven serverless pipeline across **5 AWS services** in **24 hours**, transforming real-time NOAA GFS data into global forecasts of contrail-favorable airspace
- Implemented atmospheric scoring on global **0.25°** grids, producing **~1,000 polygons** across **5 intensity tiers every 6 hours**
- Containerized Python inference with Docker; achieved **~20s** runtime per scene and autonomous operation **under $1/day**

Empathetic Help-Seeking System | Human/AI Interaction | UW | April 2026 - Present

- Built an LLM-assisted response-rewriting prototype and compared **15+** strategies across **3 models**
- Conducted and analyzed a **20-participant** study measuring perceived empathy and willingness to seek help

Snap Select | August 2025 - Present

- Developed an OCR screenshot tool with clipboard integration, achieving **~30ms** end-to-end latency
- Achieved **94%** accuracy scanning text from image types of varying quality

Imaginarium | DubHacks '25 | October 2025

- Built a real-time AI story-generation system with context-aware responses to user input in under 24 hours
- Implemented authentication, data persistence, and payment integration to support premium features